This analysis provides a detailed yet condensed overview of the core concepts of sampling methods, errors, and bias as presented in the sources, which are part of "Mathematics for Computer Science Engineers."

I. Fundamental Concepts: Samples, Bias, and Error

Sampling involves selecting a subset of units from a larger **Population**. The goal is to obtain a **Representative Sample**, which accurately reflects the characteristics of the population, thereby avoiding a **Biased Sample**.

The entire subject matter is divided into two primary categories: **Probability Samples** and **Non-Probability Samples**.

Errors in Sampling are divided into two main types:

1. **Sampling Error (Random error):** This is the discrepancy between a sample statistic and its population parameter. It occurs when the sample is simply **not representative of the population**. This error is rooted in **sampling variation**, where two different samples from the same population will naturally differ from each other.
2. **Non-sampling Error (Systematic error):** These are mistakes made during data collection or processing (e.g., data entry errors, defective questionnaires, or failure to locate the correct household).

Sampling Bias is distinct from error, occurring when the sampling technique causes the chosen sample to be non-representative. Major forms of bias include **selection bias** (some population members have a higher or lower sampling probability than others) and **non-response bias** (resulting from the absence of subjects who refuse to respond or cannot be contacted).

II. Probability Sampling Techniques

Probability sampling ensures that **every unit in the population has a probability (greater than zero) of being selected**, and this probability can be accurately determined. This type of sampling inherently decreases bias and sampling error. A key design principle is the **Equal Probability of Selection (EPS) design**, where every element has the same chance of selection, making the design 'self-weighting'.

Method	Key Mechanism & Use	Advantages	Disadvantages

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Simple Random Sampling (SRS)	**Entirely random**; each unit has an equal chance of selection. Requires a complete **sampling frame**. Best used when the population is **small**.	**Simple to use**; estimates are easy to calculate; usually representative; low sampling error.	Impracticable if the sampling frame is large; **minority subgroups** may not be sufficiently represented; lacks the use of available knowledge concerning the population.
Systematic Sampling	Relies on arranging the population and selecting elements at **regular intervals (k)**. k is the selection interval, calculated as Population Size (N) / Sample Size (n). The **first element is selected randomly**.	Sample is **evenly spread** over the entire population; cost-effective; easy to select.	**Might lead to bias** if there is an **underlying pattern/periodicity** in the population that coincides with the selection interval.
Stratified Sampling	Population is divided into **non-overlapping, internally homogeneous groups (strata)** based on shared characteristics (e.g., age, gender). **Simple random samples are drawn from *each* stratum**. Used when population proportion must be reflected in the sample.	**Enhances representativeness**; ensures adequate representation of minority subgroups; higher statistical efficiency and precision, often requiring a **smaller sample size**.	Sampling frame must be prepared separately for each stratum; **time consuming and expensive**; complex designs can lead to classification errors.
Cluster Sampling	Population is divided into **non-overlapping clusters (areas/groups)**, where each cluster is a miniature of the population (members are **heterogeneous**). **Entire clusters are randomly selected**.	Convenient for geographically dispersed populations; reduces travel costs and simplifies administration; **requires fewer resources**.	**Statistically less efficient** when cluster elements are similar; prone to **higher sampling error and biases** if clusters represent a biased opinion of the whole population.

III. Non-Probability Sampling Techniques

Non-Probability sampling is characterized by the fact that **every unit does not have a chance/probability of being selected** (or the probability cannot be accurately determined). Selection is **non-random** and based on assumptions or convenience. This method is **more likely to produce a biased sample** and restricts generalization; it also does not allow for the estimation of sampling errors.

Method	Key Mechanism & Use	Advantages	Disadvantages

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Convenience Sampling	Also known as grab or opportunity sampling. Sample elements are selected simply because they are **readily available and convenient** to the researcher.	Useful in pilot studies; **low cost and inexpensive** way to gather initial data; simple to implement.	**Prone to significant bias**; sample may not be representative of the population, limiting generalizability and **external validity**.
Judgemental/Purposive Sampling	The researcher **chooses the sample** based on who they think is appropriate, relying on their expertise or knowledge. Used when there is a **limited number of experts** in the area being researched.	Consumes minimum time; allows the researcher to use their expertise.	**Low level of reliability and high levels of bias**; prone to errors in researcher judgment; inability to generalize findings.
Quota Sampling	Population is segmented into subgroups (like stratified sampling), but units are selected based on predetermined characteristics until **pre-defined quota controls are satisfied**. **Judgment, not random selection**, is used for final selection.	Cost-effective; improves representation of certain groups within the population.	**Impossible to determine sampling error**; risk of sampling bias based on ease of access; cannot make statistical inferences leading to problems of generalization.
Snowball Sampling	Subjects are selected based on **referral from initial respondents**. Effective when the desired sample characteristic is **rare** or the sampling frame is difficult to identify (e.g., illegal immigrants or shelterless people).	Allows reaching difficult-to-sample populations via referral; cheap, simple, and cost-efficient.	Significant risk of **selection bias** (referred individuals often share common traits); the researcher has **little control** over the method; representativeness is not guaranteed.